



Giovanni De Felice

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🌐 gdefe.github.io

in giovanni-de-felice

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Languages

🇮🇹 Italian - Native

🇬🇧 English - Professional

🇫🇷 French - Basic

Short Bio

Postdoc at [Università della Svizzera Italiana](#) (CH). My research focuses on **Interpretability-by-design**: building deep learning models whose reasoning is interpretable by construction.

Experience

- 2025 - Ongoing. **Postdoctoral researcher** (Prof. S. Santini)
Università della Svizzera Italiana, Lugano, (CH) 🇨🇭
- Structured interpretable *models*.
 - *Theory* of interpretable deep learning.
 - Interpretability for deep time series models.
- 2020 - 2024 **Ph.D. candidate** in Computer Science
University of Liverpool, Liverpool (UK) 🇬🇧
- Machine learning for time series & spatio-temporal data.
- 2024 **Research visit** (Prof. C. Alippi)
Swiss AI lab IDSIA & USI, Lugano (CH) 🇨🇭
- Graph deep learning for time series.
- 2018 - 2020 **Master of Science (M.Sc.)** in Particle Physics
University of Pisa, Pisa (IT) 🇮🇹
Final grade: 110/110 cum Laude, GPA: 29.5/30
- 2020 **DOE-INFN Summer Intern** (Dr. R. H. Bernstein)
Fermilab, Batavia, IL (USA) 🇺🇸

Selected works

2025 - Ongoing. Lead author of **PyTorch Concepts**.



- Python library to fuel research on concept-based interpretable deep learning models.
- <https://pytorch-concepts.readthedocs.io/en/latest/>
- https://github.com/pyc-team/pytorch_concepts/tree/master

- **The Standard Interpretable Model: A general theory of interpretable machine learning to deductively design interpretable methods using Lagrangian mechanics**
P Barbiero, **G. De Felice**, M. Zarlenga, F. Giannini, F. Bonchi, M. Jamnik, G. Marra, R. Noris.
Under submission. *arXiv preprint arXiv:2606.12289*. 2026.
- **Interpretability in Deep Time Series Models Demands Semantic Alignment**
G. De Felice, R. D'Elia, A. Termine, P. Barbiero, G. Marra, S. Santini.
International Conference of Machine Learning (ICML). Position paper track. 2026.
- **Mixture of Concept Bottleneck Experts**
F. De Santis, G. Ciravegna, **G. De Felice**, A. Casanova, F. Giannini, M. Diligenti, M. Zarlenga, P. Barbiero, J. Schneider, D. Giordano.
International Conference of Machine Learning (ICML). **Spotlight (Top 2.2%)**. 2026.
- **Causally Reliable Concept Bottleneck Models**
G. De Felice, A. Casanova, F. De Santis, S. Santini, J. Schneider, P. Barbiero, A. Termine.
Advances in Neural Information Processing Systems (NeurIPS). 2025.

Talks & dissemination

- **Invited talk** at **ETH Zurich** (Mar 2026, “PyTorch Concepts: Towards Standard Interpretable Machine Learning”.
- **Guest lecture** at **Università della Svizzera Italiana** (Mar 2026, “Standard Interpretable Machine Learning”, in “Advanced Topics in Machine Learning” (Prof. Cesare Alippi).
- **Invited talk** at **University of Cambridge** (Feb 2026, “Towards Causally Reliable Concept-based Models”.
- **Invited talk** at **KU Leuven** (Feb 2026, “PyTorch Concepts: interpretable and causally transparent deep learning models.”
- **Invited talk** at ‘*Temporal Graph Reading Group*’ (Mar 2024, Virtual).
Presentation of the paper: “Graph-based Virtual Sensing from Sparse and Partial Multivariate Observations”.
- **Invited talk** at **Beckers group** (Nov 2023, Invited to present my work to the Executive Management team of the Beckers Group.
- **Invited talks** at two *Mu2e Collaboration Meetings* (Oct 2020,  and Jun 2020, Virtual).
- **Poster presentations** at PGR Workshop (Univ. of Liverpool), NeurIPS 2022 (New Orleans), NeurIPS 2023 (New Orleans), ICLR 2024 (Vienna), NeurIPS 2025 (San Diego).

Program committee member

- **Reviewer** for *International Conference on Neural Information Processing Systems* (NeurIPS)
- **Reviewer** for *International Conference on Machine Learning* (ICML) - **Gold Reviewer** award (top 25%)
- **Reviewer** for *International Conference on Learning Representations* (ICLR)
- **Reviewer** for *IEEE Transactions on Neural Networks and Learning Systems* (TNNLS) journal

Industrial projects

2020 - 2024 **Weathering Predictions of Paint Formulations**, in partnership with **Beckers Group**.



Beckers Group / University of Liverpool

- Predict long-term performances in untested locations from climatic data.
- Extract formulatory information from data.

Musical studies

I started playing the **classical piano** when I was 8 yo, all the way to studying at the conservatory. It now constitutes a central part of my life.

- Classical Piano Degree Admission (8th/10 year exam)
Higher institute of musical studies Pietro Mascagni, Livorno, Italy. Grade: 8/10 (**2nd place**) (2015)
- Mid-term exam (5th/10 year exam)
Conservatory Alfredo Casella, l'Aquila, Italy. Grade: 9/10 (2013)
- Solfeggio and theory of music
Conservatory Nino Rota, Monopoli, Bari, Italy (2011)
- GRADE 1 exam
The Associated Board of the Royal Schools of Music, Varese, Italy. Grade: pass with distinction (2007)
- Other: Alto Saxophone (2022-Ong.), keyboard live concerts in Italy and France (2013-2015), pianist in a theater-dance spectacle (2014), orchestra and chamber music (2010-2015).

All publications

Machine Learning:

- **The Standard Interpretable Model: A general theory of interpretable machine learning to deductively design interpretable methods using Lagrangian mechanics**
P Barbiero, **G. De Felice**, M. Zarlenga, F. Giannini, F. Bonchi, M. Jamnik, G. Marra, R. Noris
arXiv preprint arXiv:2606.12289. 2026.
- **A foundation model for electrodermal activity data**
L. Alchieri, M. Garzon, L. Alecci, F. B. De Bona, M. Gjoreski, **G. De Felice**, S. Santini
arXiv preprint arXiv:2603.16878. 2026
- **Interpretability in Deep Time Series Models Demands Semantic Alignment**
G. De Felice, R. D'Elia, A. Termine, P. Barbiero, G. Marra, S. Santini.
International Conference of Machine Learning (ICML). 2026
- **Mixture of Concept Bottleneck Experts**
F. De Santis, G. Ciravegna, **G. De Felice**, A. Casanova, F. Giannini, M. Diligenti, M. Zarlenga, P. Barbiero, J. Schneider, D. Giordano
International Conference of Machine Learning (ICML). **Spotlight (Top 2.2%)**. 2026.
- **Federated Concept-Based Models: Interpretable models with distributed supervision**
D. Fenoglio, A. Casanova, F. De Santis, M. Li, G. Dominici, J. Schneider, M. Gjoreski, M. Langheinrich, P. Barbiero, **G. De Felice**
arXiv preprint arXiv:2602.04093. 2026
- **Causally Reliable Concept Bottleneck Models**
G. De Felice, A. Casanova, F. De Santis, S. Santini, J. Schneider, P. Barbiero, A. Termine.
Advances in Neural Information Processing Systems (NeurIPS). 2025
- **On the Regularization of Learnable Embeddings for Time Series Processing**
L. Butera, **G. De Felice**, A. Cini, C. Alippi.
Transactions on Machine Learning Research (TMLR). 2025.
- **Graph-based Virtual Sensing from Sparse and Partial Multivariate Observations**
G. De Felice, A. Cini, D. Zambon, V. Gusev, C. Alippi.
International Conference on Learning Representations (ICLR). 2024.

- **Time Series Kernels based on Nonlinear Vector AutoRegressive Delay Embeddings**
G. De Felice, J. Y. Goulermas, and V. Gusev.
Advances in Neural Information Processing Systems (NeurIPS). 2023.
- **Spatio-Temporal Weathering Predictions in the Sparse Data Regime with Gaussian Processes**
G. De Felice, V. Gusev, J. Y. Goulermas, M. Gaultois, M. Rosseinsky, C. Gauvin,
AI for Science: Progress and Promises (NeurIPS Workshop). 2022.

Physics:

- **Mu2e Run I Sensitivity Projections for the Neutrinoless $\mu^- \rightarrow e^-$ Conversion Search in Aluminum**
Mu2e Collaboration,
Universe, 9(1), p.54. 2023.
- **An updated estimate of the Mu2e experiment sensitivity**
G. De Felice,
Master's degree thesis. 2020.

Materials Informatics:

- **Enhancing Extrapolation in Materials Science through Contrastive Learning of Chemical Compositions**
O. Federico, G. De Felice, R. Savani, V. Gusev, and M. Rosseinsky.
AI for Accelerated Materials Design (NeurIPS Workshop). **Spotlight**. 2023.
- **Not as simple as we thought: a rigorous examination of data aggregation in materials informatics**
O. Federico, G. De Felice, V. Gusev, and T. Sparks.
Digital Discovery, 3 (2), p.337-346. 2023.